A COVERING YIELDS AN EXACT PRESENTATION \BIGOPLUS J_{U_I \TIMES_U U_J!}\MATHCAL{O} \TO \BIGOPLUS J_{U_I!}\MATHCAL{O}_{U_I} \TO J_!\MATHCAL{O}_U \TO 0

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one displayed equation of the same chapter and one Homology lemma, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Maps out of j_!\mathcal{O}_U into a module. Stacks Project, Modules on Sites, displayed equation equation-map-lower-shriek-OU-into-module. Supplies the identification of \Hom(j_!\mathcal{O}_U, \mathcal{F}) and the two maps with \mathcal{F}(U) \to \prod \mathcal{F}(U_i) \to \prod \mathcal{F}(U_i \times_U U_j).
- Exactness checked after applying Hom. Stacks Project, Homology, Lemma homology-lemma-check-exactness. Supplies the criterion by which exactness of the Hom-into-(-, \mathcal{F}) sequences for all \mathcal{F} yields exactness of the original sequence.

Source reference. Original equation 0G1V: sources/stacks-chow/tags.txt, source lines 20035–20035. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 05AA: sources/stacks-chow/tags.txt, source lines 6843–6843. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\{U_i \to U\}$ be a covering of $\mathcal{C}$. Then the sequence*

$$\bigoplus j_{U_i \times_U U_j !}\mathcal{O}_{U_i \times_U U_j} \to \bigoplus j_{U_i !}\mathcal{O}_{U_i} \to j_!\mathcal{O}_U \to 0$$

is exact.

Proof. For any $\mathcal{O}$-module $\mathcal{F}$ the functor $\mathrm{Hom}_{\mathcal{O}}(-, \mathcal{F})$ turns our sequence into the exact sequence $0 \to \mathcal{F}(U) \to \prod \mathcal{F}(U_i) \to \prod \mathcal{F}(U_i \times_U U_j)$, see (0G1V). The lemma follows from this and Homology, Lemma 05AA. □

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